

Retail Sales Analytics: Customer Behavior and Revenue Driver Analysis



Adin Nugroho



adinn41@gmail.com



github.com/adn30/Portofolio

Executive Summary

Project

Background

A retail company wanted to understand:

customer **purchasing behavior, factors driving sales,** and **data-driven revenue growth** opportunities.

a 2023 retail sales dataset was analyzed to identify customer patterns, time trends, and the contribution of product categories to business performance.

Tools

- Python (Pandas, NumPy) for main analysis



Objective & Tools

Objective

Identify **key factors influencing** sales performance and customer behavior to support data-driven business strategies.

Metrics

- Total Revenue
- Average Spending per Customer
- Purchase Frequency
- Quantity per Transaction
- Price per Unit
- Product Category Contribution
- Time-based Sales Trend

Analysis Method

- Distribution Analysis
- Time Series Analysis
- Customer Segmentation
- Correlation Analysis
- Transaction Behavior Analysis

Results

Customer Behavior

- **Female** customers have the **highest shopping frequency** and revenue contribution.
- **Gender influences product preferences** more than age.
- **Age influences purchasing power** but does **not significantly** determine purchasing patterns.

Product Performance

- **Clothing & Electronics** product category is the **main contributor to revenue**.
- **Beauty** product category has a **high average price**, which **increases average spending**.

Time-Based Sales Pattern

- The **highest** sales occurred in **May (\$53,150)**.
- The **lowest** was in **September (\$23,620)**.
- Transactions **peaked at the end of the week** (Friday–Saturday).
- **The May spike** was driven by electronics sales (indicating a campaign/promotion).

Recommendation

1. Dual Revenue Strategy

Separate product sales strategies
Value Driver → Electronics & Beauty (high price)
Volume Driver → Clothing (high quantity)

2. Gender-Based Marketing

Gender is a stronger determinant of product preference.
Female → Beauty and Clothing boost campaign
Male → Electronics special promotion

3. Time Optimized Promotion

optimize main promo for **weekend and seasonal campaign**

4. Basket Expansion Strategy

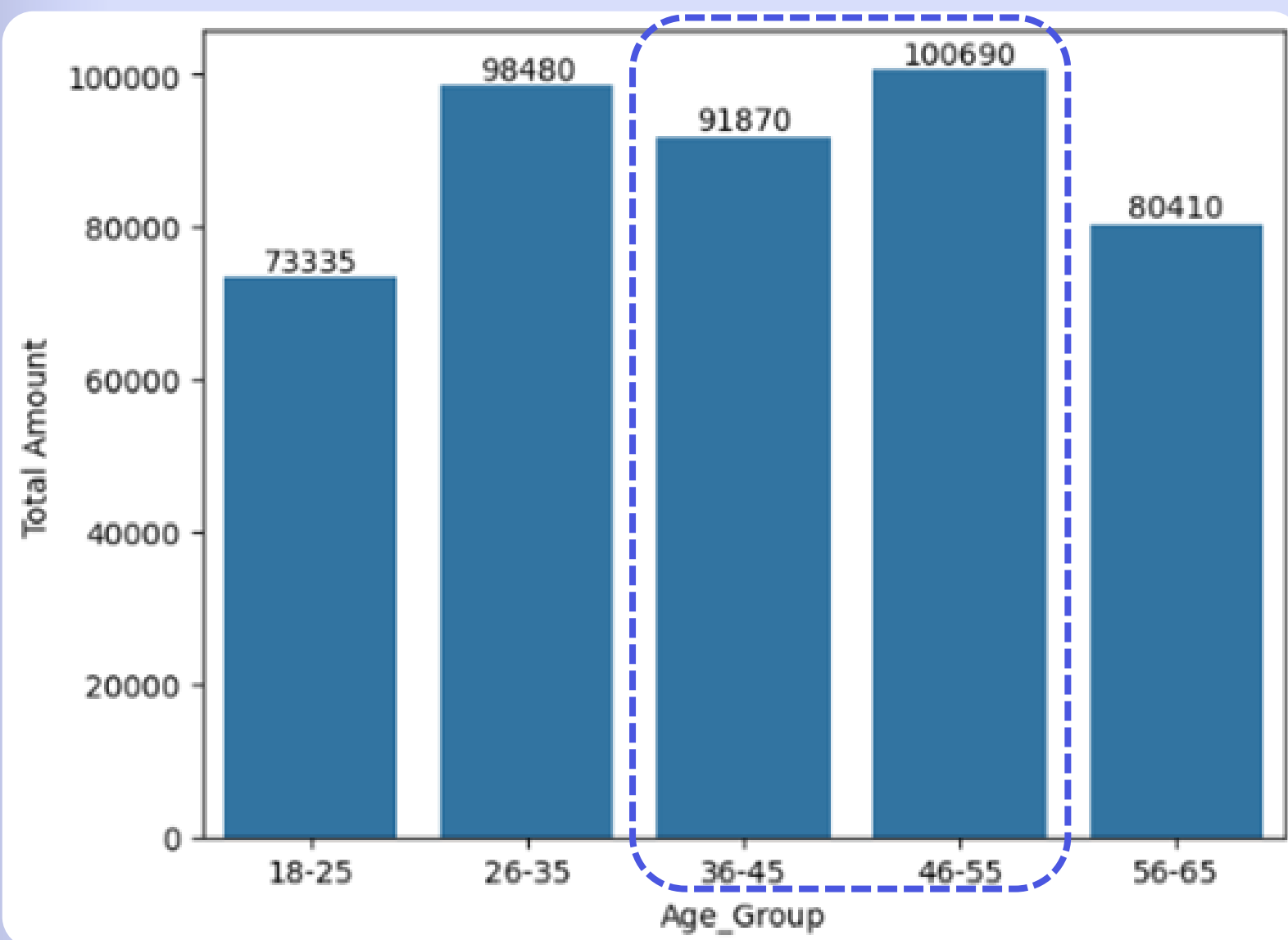
Encourage multi-item purchases through bundling and cross-selling product **to boost revenue** from other products to be distributed evenly.

5. Revenue Growth Focus

shift focus from **increasing the volume of transaction** to **increasing the average transaction value**

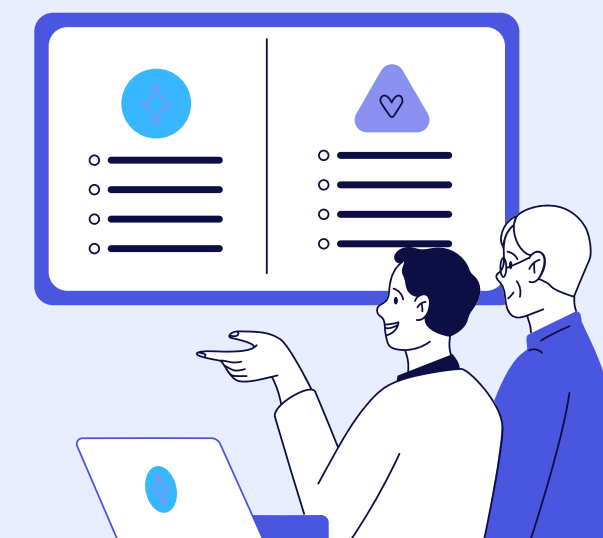
How does Customer Age and Gender Influence Purchasing Behavior?





Finding 1 – Spending Distribution by Age

In total sales, middle aged customers (36 – 55) contribute the highest spending compared to other age groups. However, **this isn't enough to answer and conclude customer spending habits based on age;** let's explore this further.

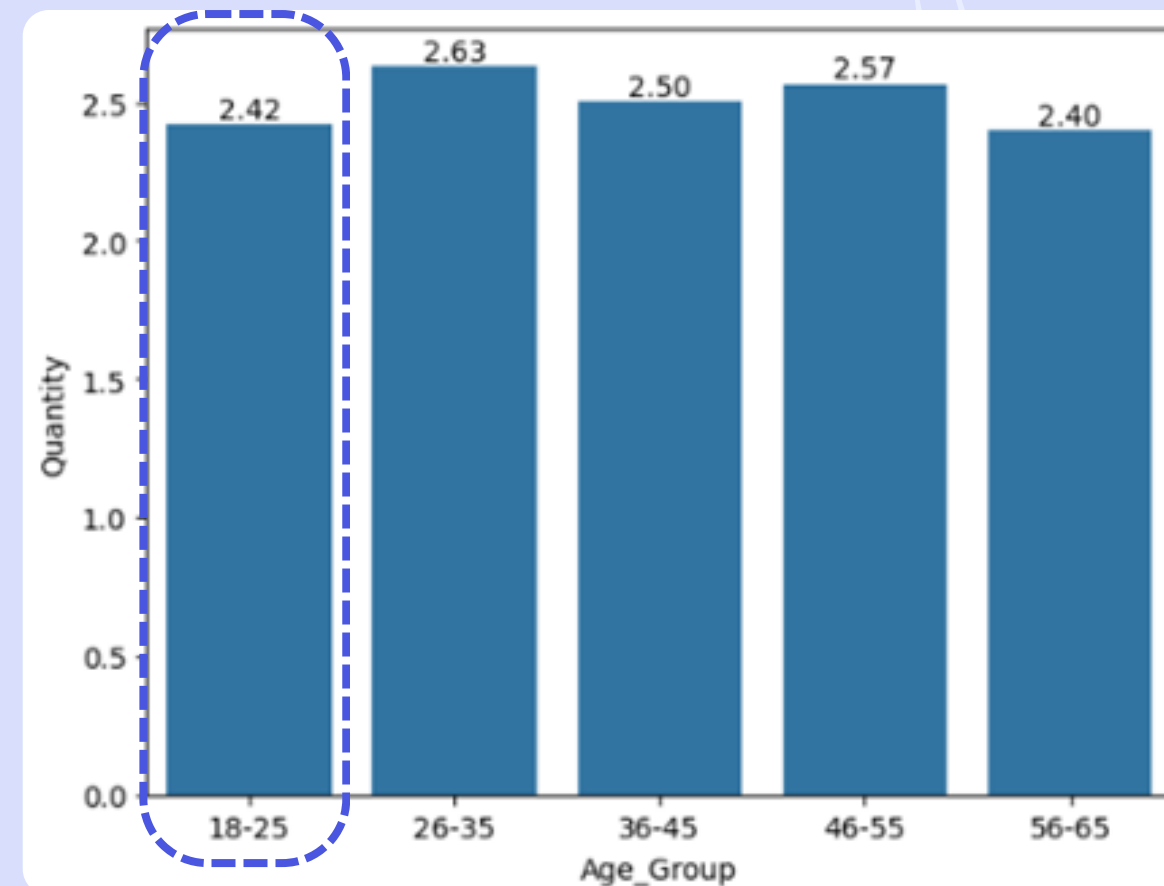




Finding 2 – Purchase Frequency



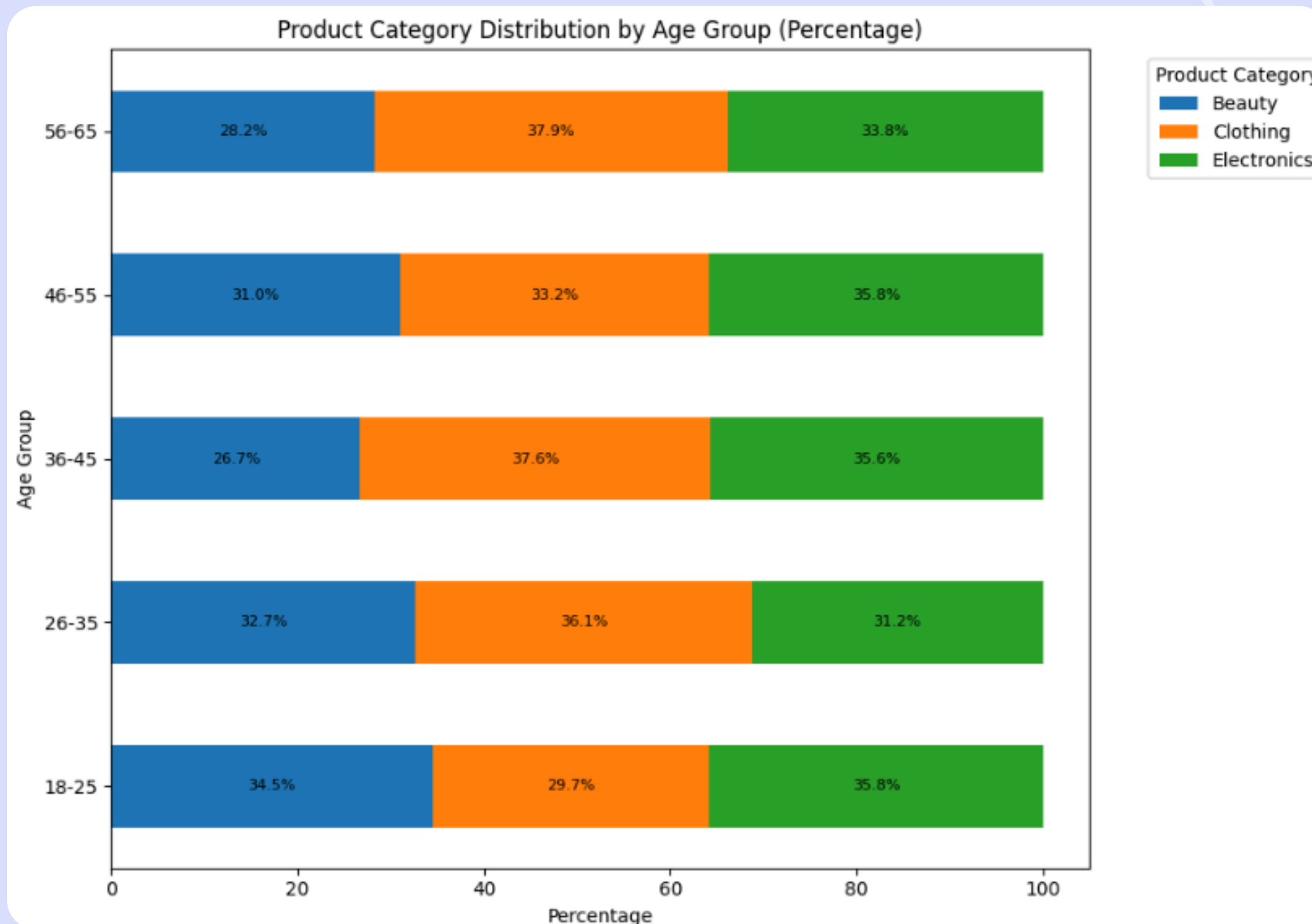
Age Group	Purchase Frequency
18 – 25	148
26 – 35	205
36 – 45	202
46 – 55	229
56 – 65	195



Purchase intensity of customers in **younger age** range **slightly fewer than others**. This occurs because their shopping frequency and the average number of items purchased are in small amount.



Finding 3 – Product Preference by Age



There is **no strong product category dominance** across age groups, suggesting age effects spending more volume than product choice

Conclusion

Customer age **influences spending capacity**, but shows limited impact on product preference patterns.



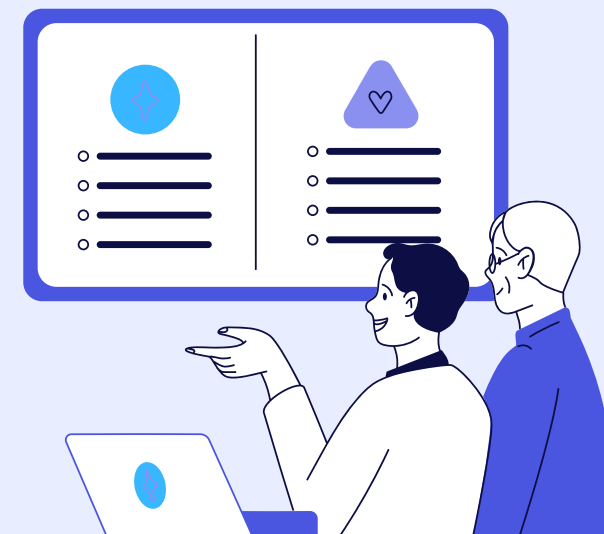
Female
510
transactions



Male
490
transactions

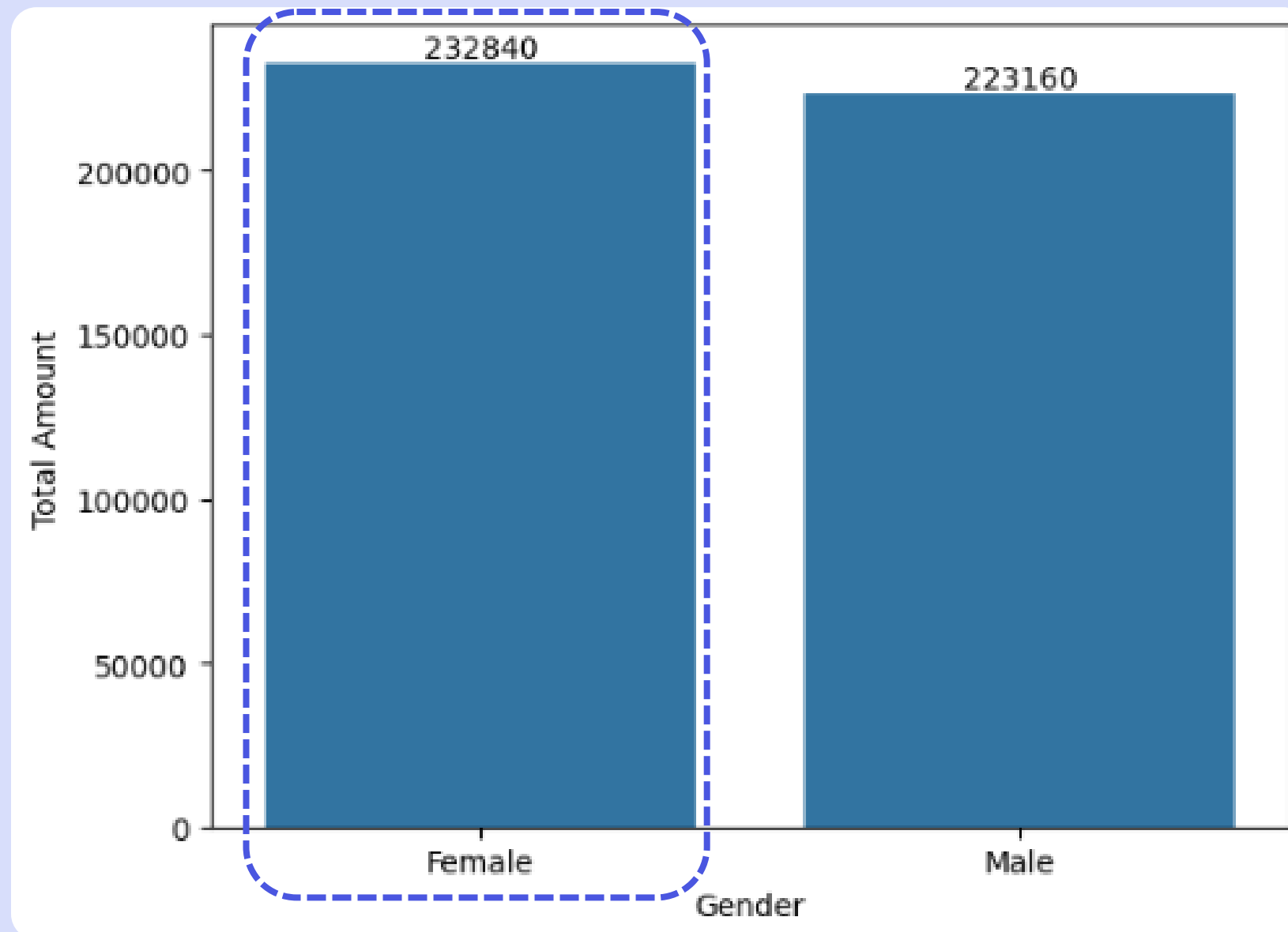
Finding 1 – Purchase Frequency

Although the difference is small, it shows consistent results that **female customers occupy the top position** in terms of shopping frequency, this shows that female customers have a greater tendency to shop than male.





Section B – Gender Influence



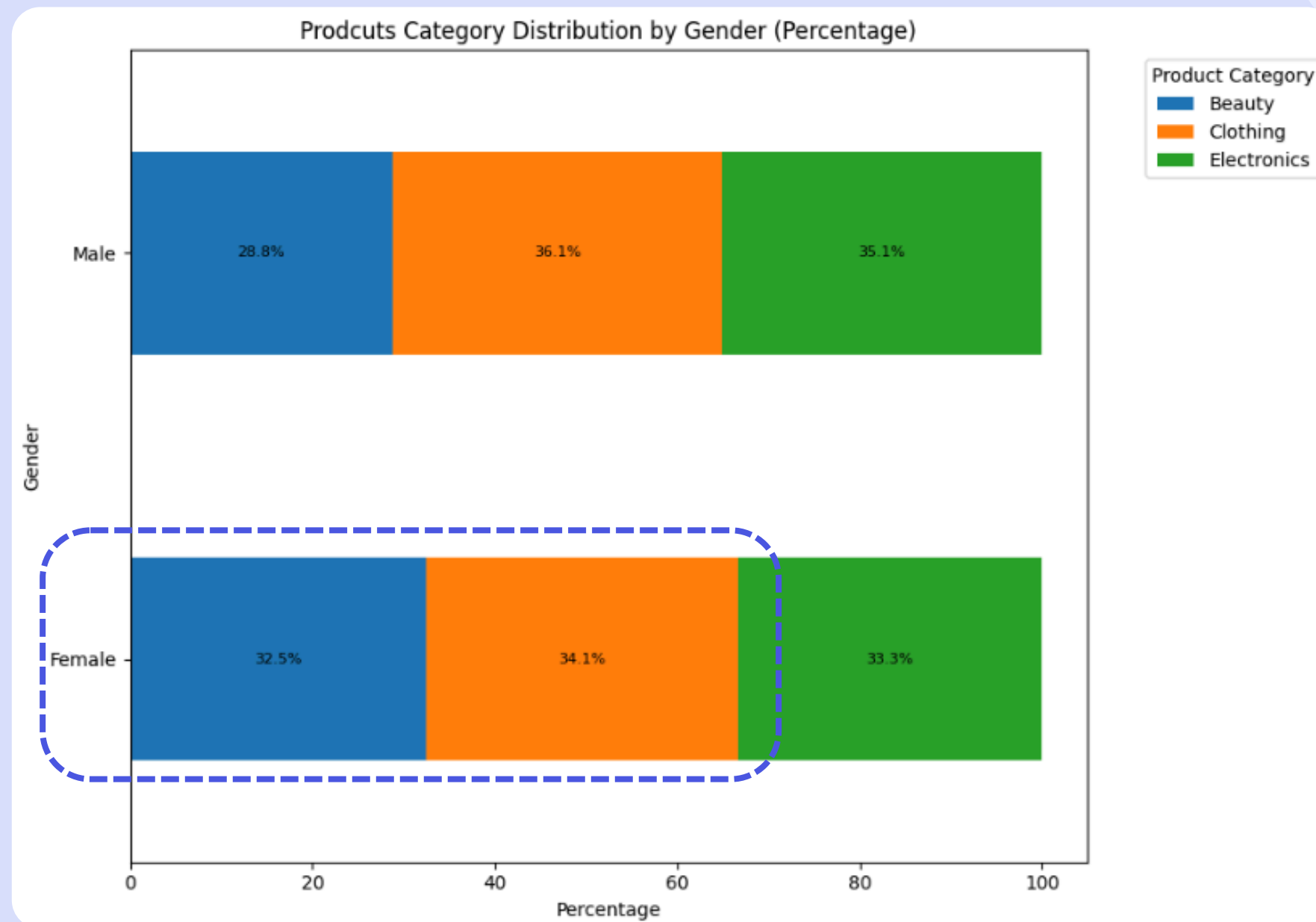
Finding 2 – Spending Comparison

There is no significant difference between the spending of female and male customers, but it is **still consistently seen that female customers have a larger total spending** than male customers.





Finding 3 – Category Preference



Female customers **dominate 2 product categories** consisting of beauty and clothing with a large percentage of **beauty products of 32.5% and clothing 34.1%.**

This shows that shopping behavior based on gender are **based on purchasing tendencies in certain categories rather than the amount of spending.**

Conclusion

Gender influences **product selection behavior more strongly** than spending magnitude



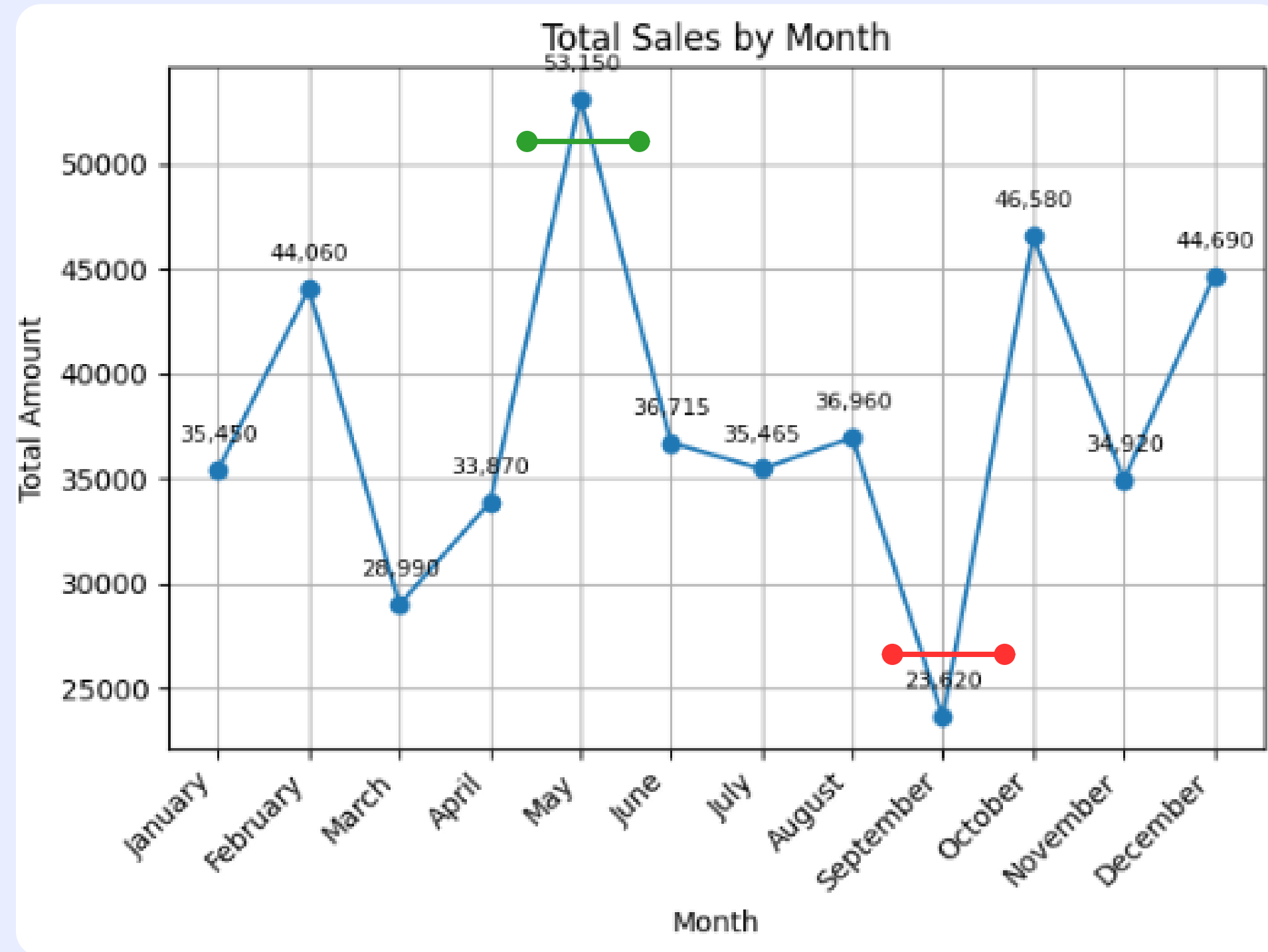
Conclusion

While age primarily affects purchasing capacity, **gender plays a stronger role in shaping product preferences.** This indicates that demographic-driven marketing strategies **should differentiate between age-based spending potential and gender-based product targeting**

**Are there discernible
patterns in sales
across different time
periods?**



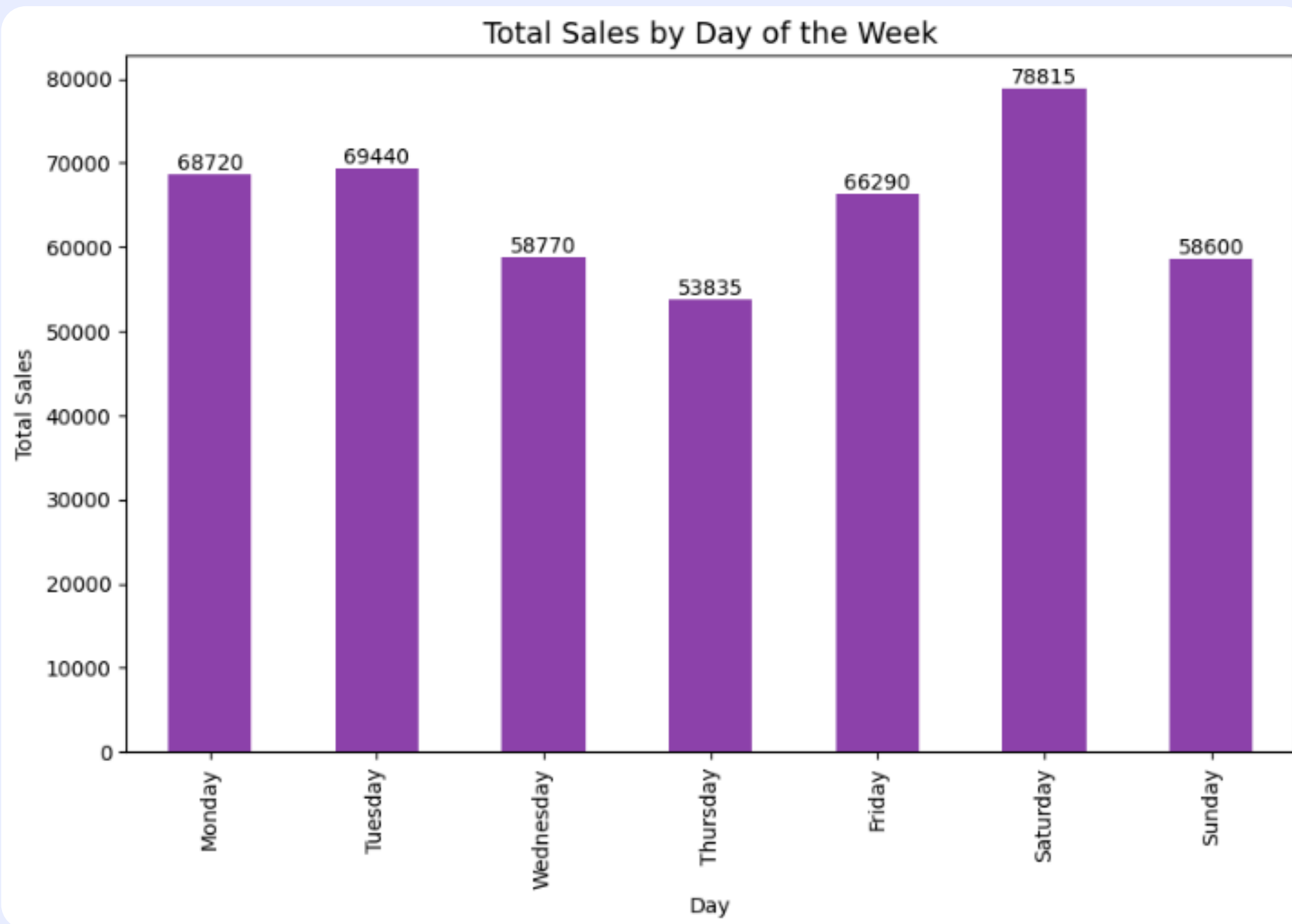
Finding 1 – Monthly Sales Performance



May was the **highest sales total of 2023**, with revenue at \$53,150. The **lowest sales total was recorded in September**, at \$23,620, indicating a significant decline in customer purchasing power following the summer holidays.

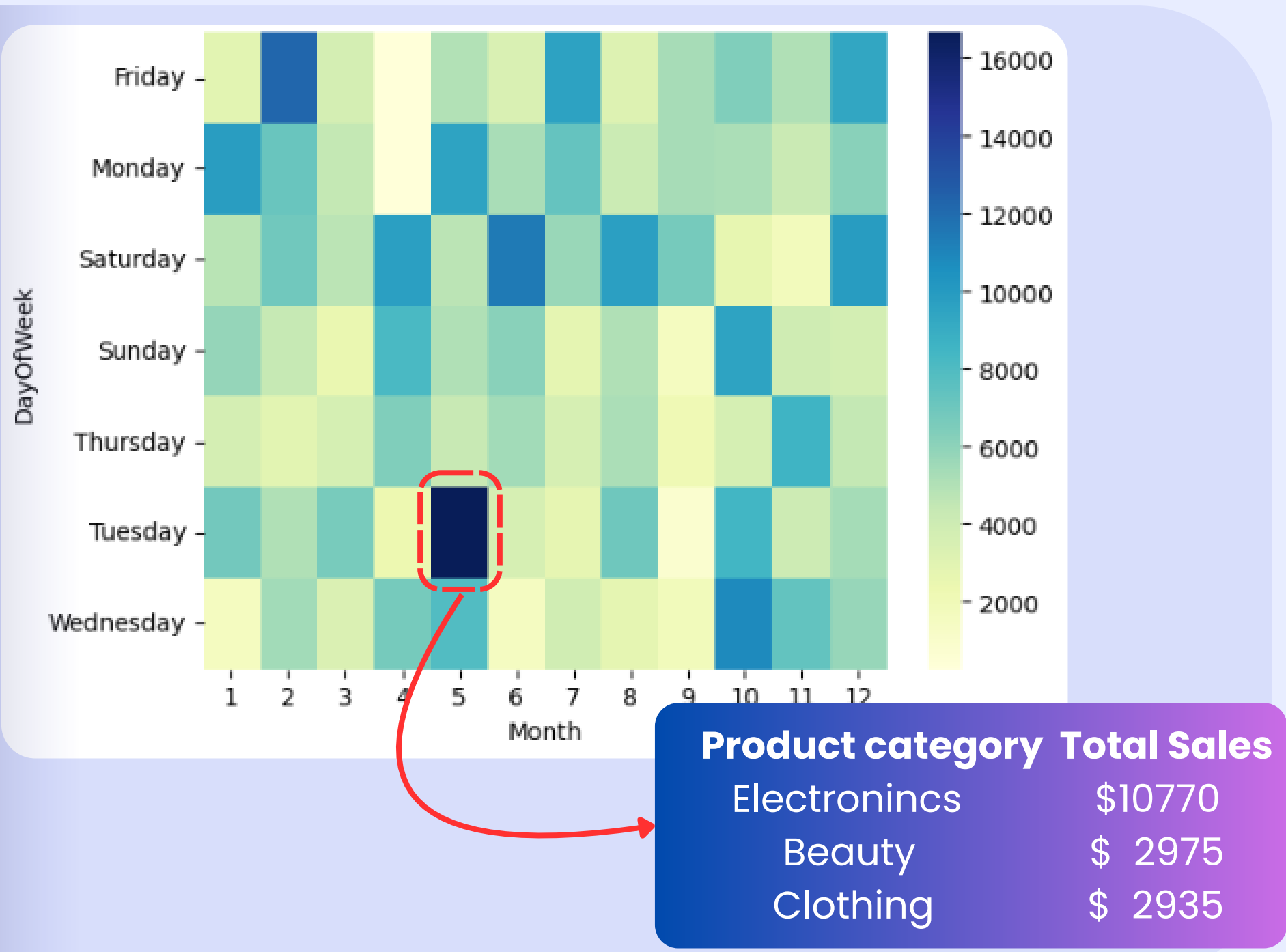
Based on the monthly sales totals, **there is a clear trend of sales fluctuations during certain periods**, such as New Year's (January-February), summer holidays (May-August), and Christmas & New Year's holidays (October-December). Let's examine the sales patterns by day of the week.

Finding 2 – Weekly Sales Performance



Customers tend to shop more **at the end of the week**, specifically on Fridays and Saturdays, where total revenue tends to increase. However, sales tend to decline mid-week, such as on Wednesdays and Thursdays.

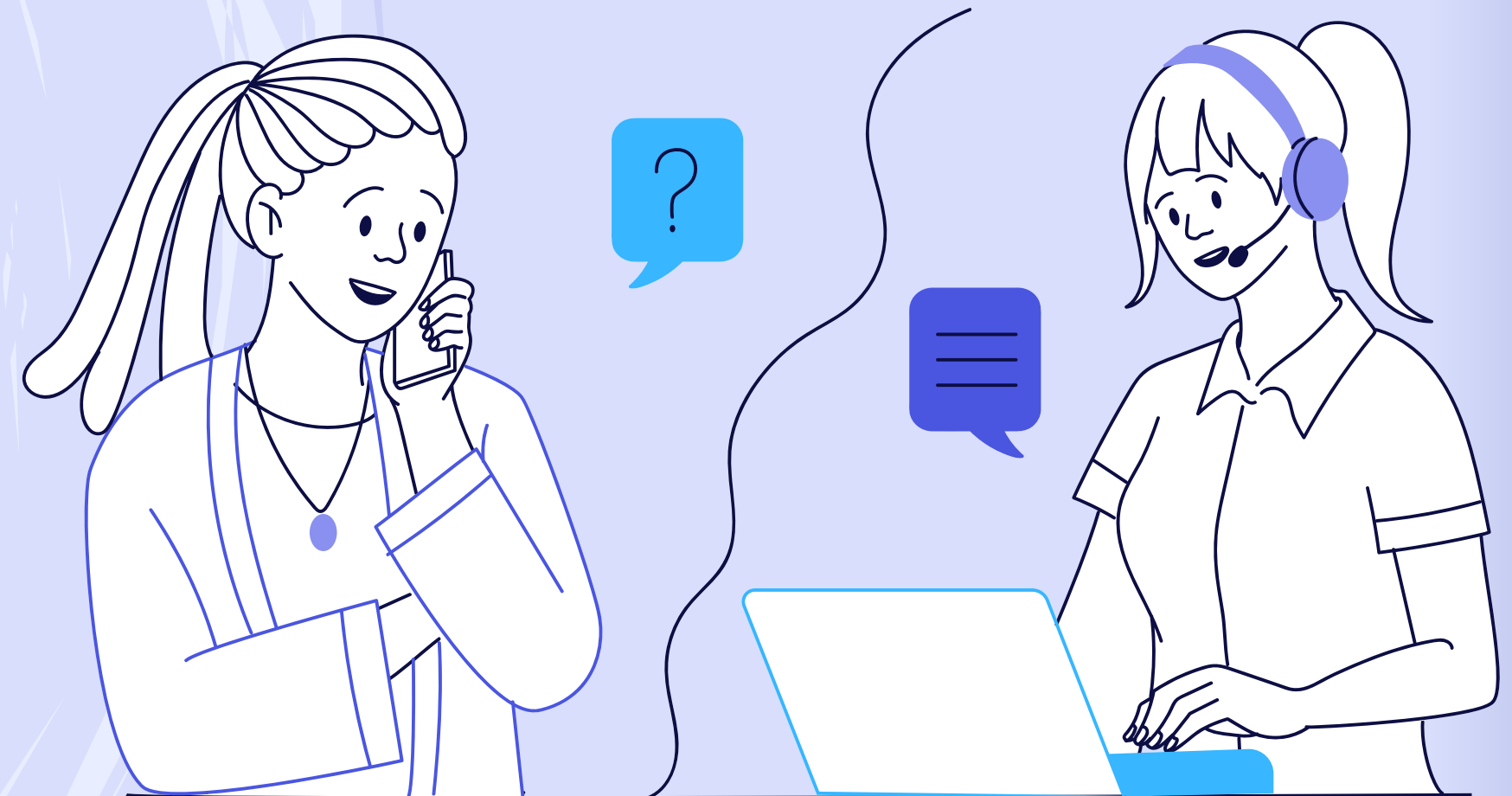
Finding 3 – Deep dive, sales anomalies in may 2023



The abnormal upward trend in the sales performance graph that occurred in May 2023 makes this an interesting discussion. Let's explore which day was the 'peak' for total sales in May.

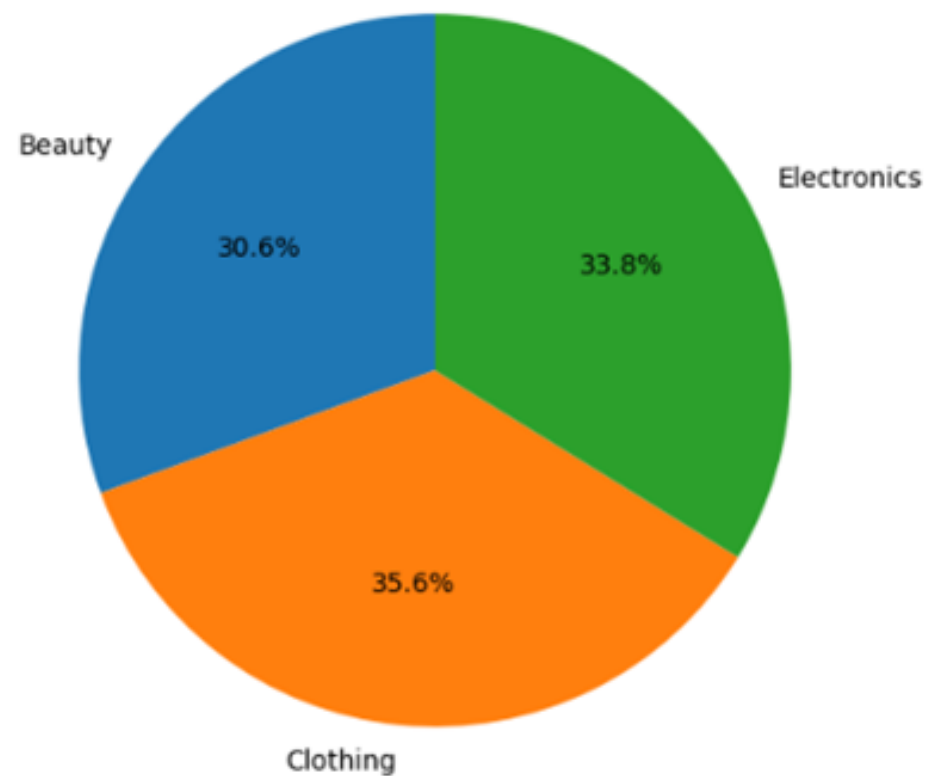
The highest total sales occurred on Tuesday, reaching \$16,680. This figure was driven by electronics purchases, which accounted for the largest portion of sales. This indicates a special sales program being implemented by the store to boost electronics sales.

Which Product categories hold the highest appeal among customers?

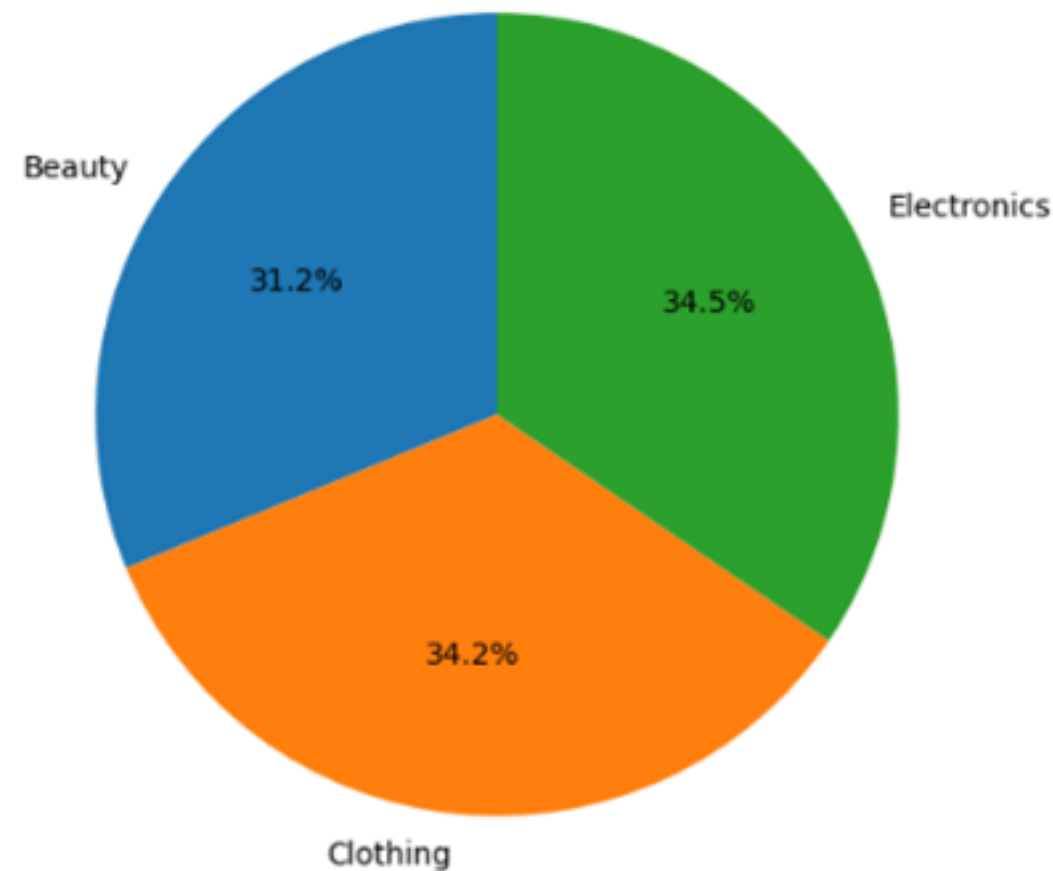


these two categories were found to be customers favorites

Percentage Quantity by Product Category



Percentage Total Amount by Product Category

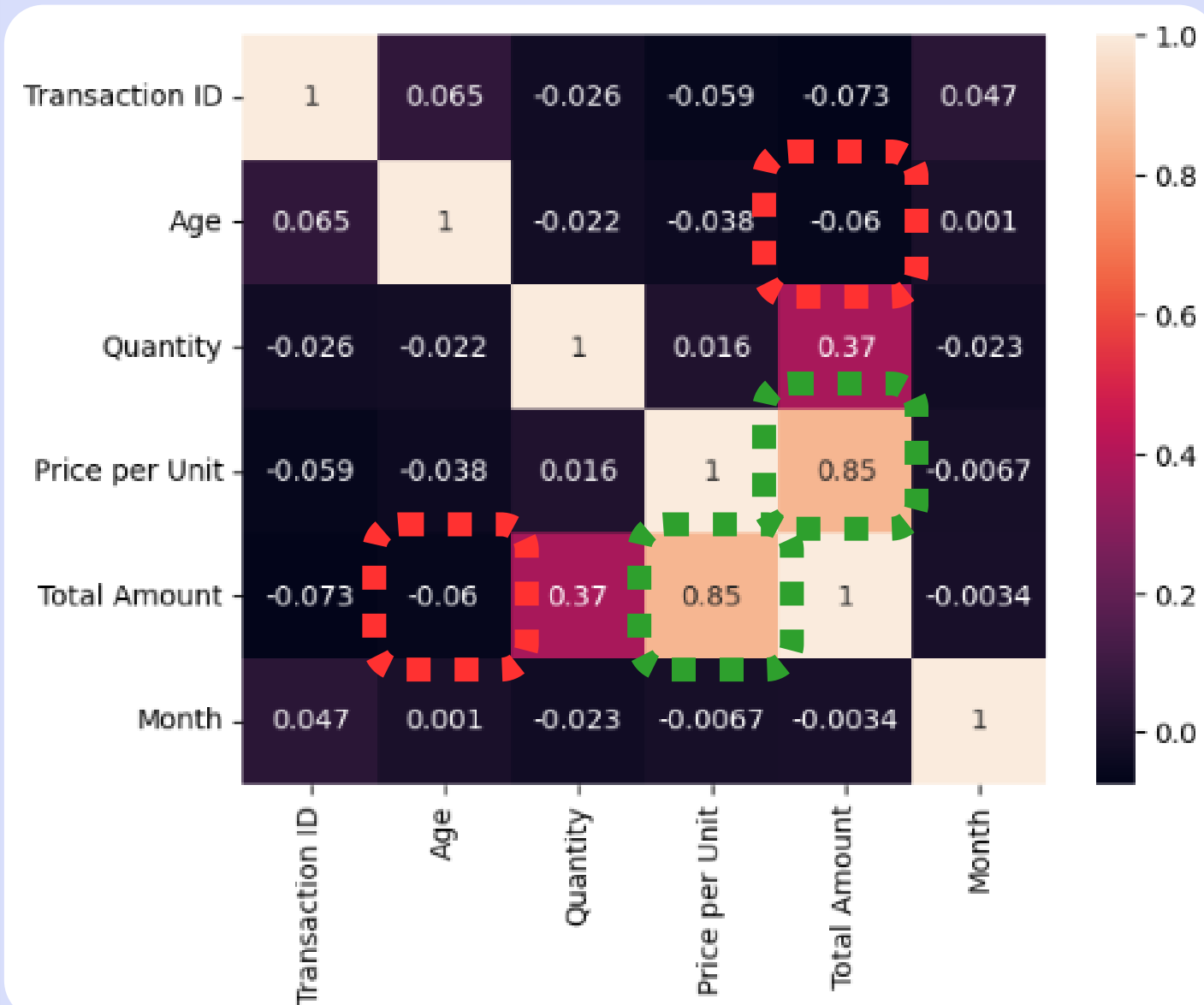


Clothing and electronics were the most popular product categories among customers, based on purchase volume and total sales, although at a glance, the comparisons were not significantly different. This indicates that these two categories were the **main drivers of total revenue in 2023.**

**What are relationships
between age,
spending, and price?**



age, spending, price which the main driver?



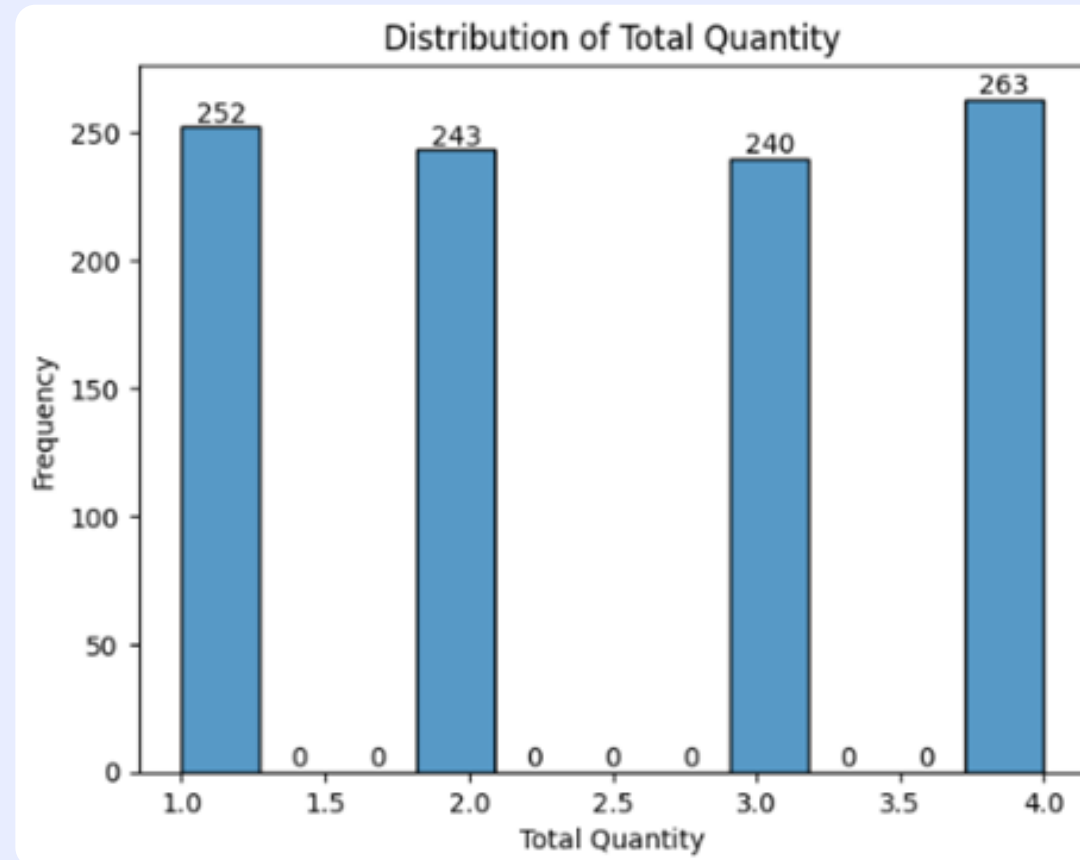
Age vs. total amount shows a **negative correlation (-0.06)** which indicates that customer age does not determine purchasing power. **Based on the previous discussion**, the purchasing power of customers with different age ranges is quite fluctuating so this cannot be concluded.

Price per unit vs. total amount shows a **positive correlation (0.85)** which indicates that **revenue is driven by the value of the goods purchased** (for example, electronic products have a high price) rather than the quantity of the goods purchased.

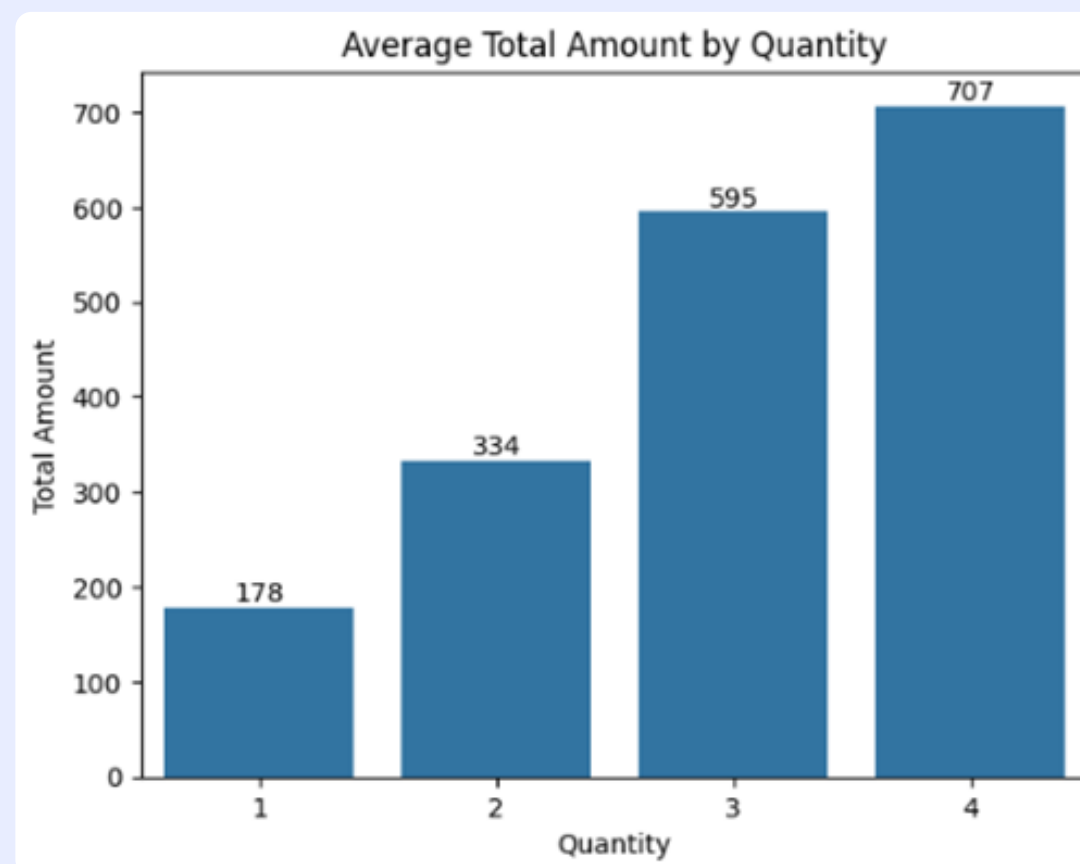
**are there distinct
purchasing behaviors on
the number of items
bought per transactions?**



Finding 1 – Purchase Quantity Distribution

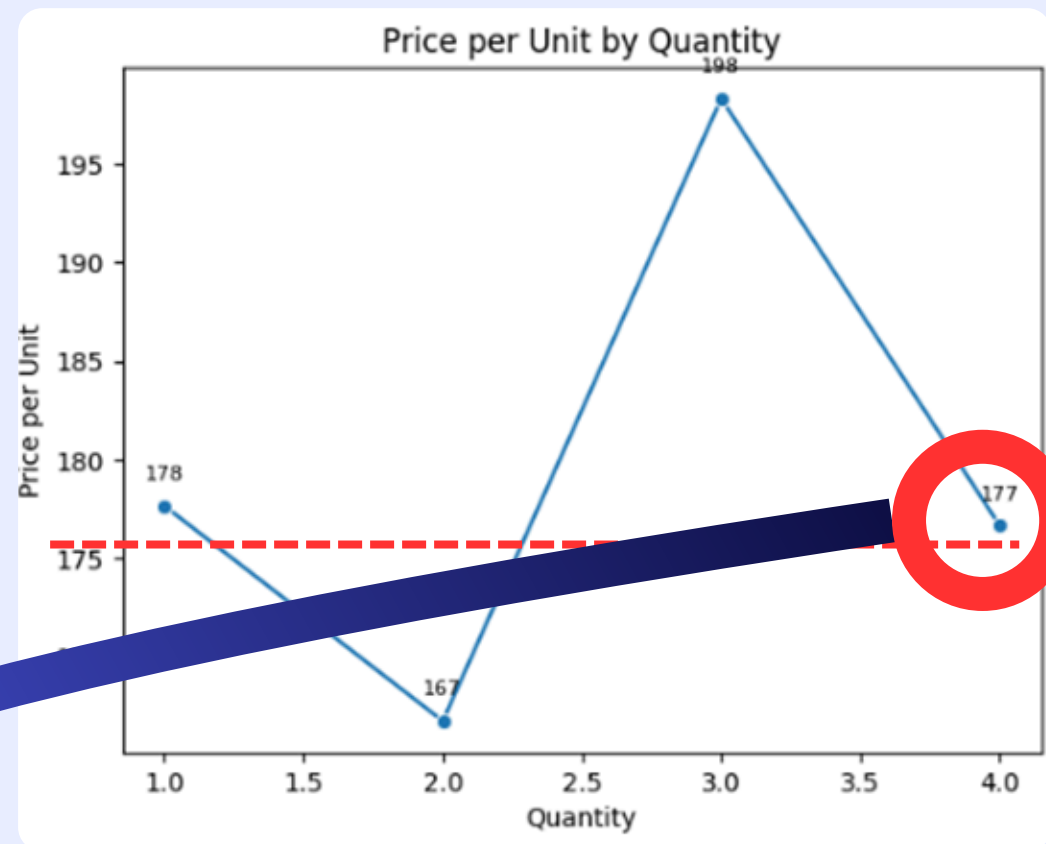


Based on frequency distribution of product purchase quantities, amount of quantity from 1 to 4 is **relatively evenly distributed, with no dominant item**. This suggests that differences in customer habits **are not due to quantity purchased, but rather to nature of transactions**.

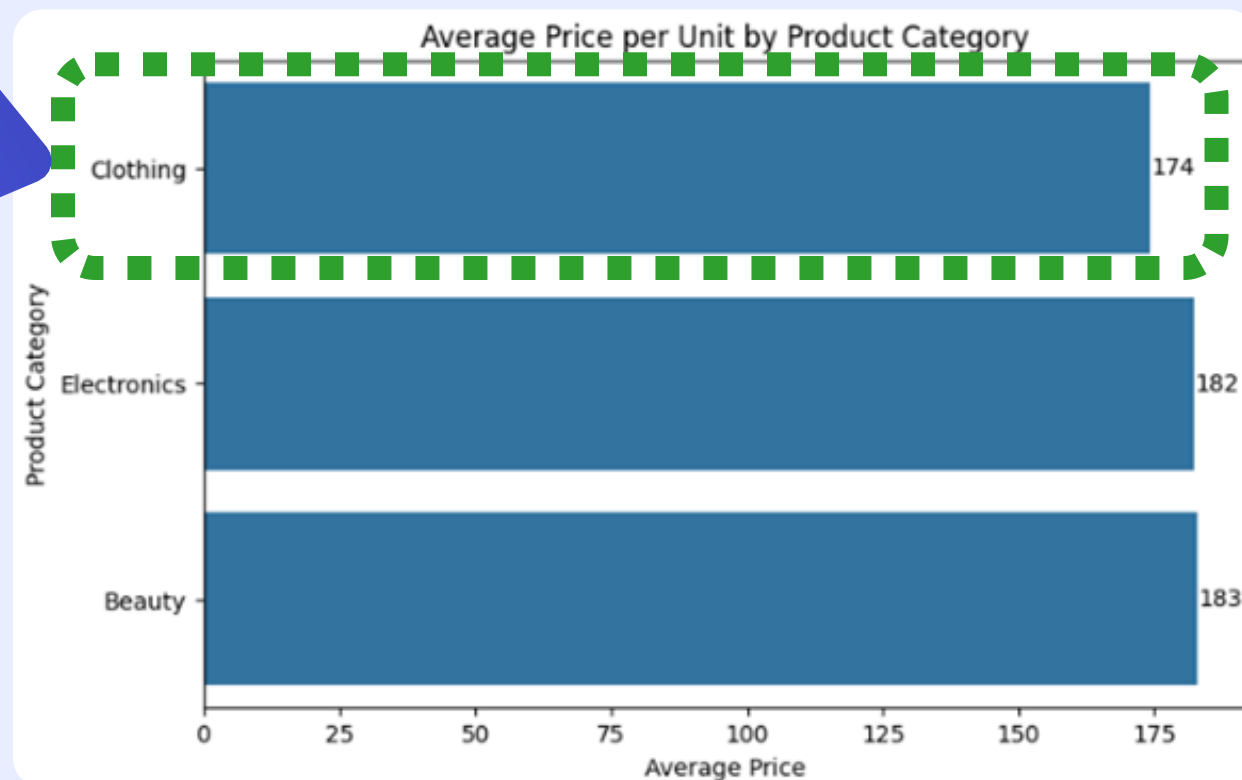


The more items purchased, the greater the average total revenue. Customers with a purchase quantity of 4 had the highest average total revenue, indicating that customers with the **largest baskets generated high-value transactions**.

Finding 2 – Which Product Category Contribute High Volume Transactions?



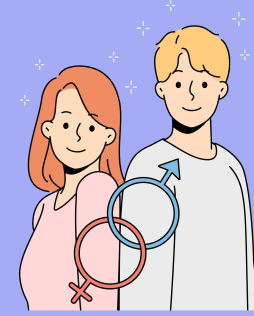
Purchases of high quantities tend to occur for lower-priced products. **We found a pattern of consumer behavior when encountering relatively low product prices:** the lower the price of a product, the higher quantity of the product purchased.



Product category with the **lowest price** is **clothing** where the average price per unit is \$174, many customers buy clothing products in large quantities.

Conclusion

Retail sales analysis shows that sales performance is not driven by a single factor, but rather a combination of:



demographic
segmentation
(gender & age)



product value
(price per unit)



purchase
momentum



transaction
characteristics
(basket behavior)

The most important insight:

Revenue is more influenced by the value of the products purchased than the quantity of items purchased.

This means:

A growth strategy **cannot simply focus on increasing transaction volume** but must **optimize the combination** of product category, timing, and target customers.



Thank You

Click for detail

[Dataset – Retail Sales](#)

[EDA With Python](#)

